DRUGS · EAR PREPARATIONS

# Antifungal ear preparations

Otomycosis is a fungal otitis externa caused mainly by Aspergillus (black or grey spores on a fluffy mat) and Candida (white, creamy debris). It follows prolonged antibiotic drops, humidity, hearing aids and immunosuppression. Cleaning the canal is as important as the antifungal; clotrimazole solution is the usual first-line drug.

## 1 CHAPTER ONE At a glance

The drugs on one page.

### 1.1 Summary

ANTIFUNGAL EAR PREPARATIONS

| | SUMMARY |
|---|---|
| **Preparations** | Clotrimazole 1% solution (first-line); miconazole or econazole creams; boric acid powder or solution; ciclopirox (where available); systemic itraconazole or voriconazole for invasive disease |
| **How they work** | Azoles block ergosterol synthesis in the fungal membrane (lanosterol 14-alpha-demethylase); boric acid acidifies and is fungistatic |
| **ENT uses** | Otomycosis; fungal infection in mastoid cavities; prophylaxis in patients prone to recurrence |
| **Dose** | Clotrimazole 2 to 3 drops two or three times daily, continued for at least 14 days after clinical resolution |
| **Pregnancy** | Topical use acceptable |

## 2 CHAPTER TWO How they work

Mechanism and handling.

### 2.1 Mechanism

Azoles deplete ergosterol and accumulate toxic sterols in the fungal membrane. They are active against Candida and Aspergillus, although Aspergillus is less sensitive to clotrimazole than to the newer triazoles.

## 3 CHAPTER THREE Using them in ENT

Indications and doses.

### 3.1 Treating otomycosis

- **Clean the canal** under the microscope: remove the fungal debris and spores by microsuction, repeated at intervals. Drops alone into a canal full of hyphae fail.
- **Clotrimazole 1%** drops for at least two weeks after the canal looks clean; stop the antibacterial drops that caused it.
- Keep the ear **dry**; treat underlying eczema; control diabetes.
- **With a perforation**, clotrimazole has not been shown to be ototoxic and is commonly used; avoid gentian violet and alcohol-based preparations, which are irritant and potentially ototoxic.

### 3.2 When to think further

Persistent or recurrent fungal infection in a diabetic or immunocompromised patient, with pain and granulation, may be **invasive fungal necrotising otitis externa** (usually Aspergillus): biopsy and treat systemically with voriconazole.

# 4 CHAPTER FOUR Safety

What goes wrong.

## 4.1 Adverse effects

Local irritation and burning; rarely contact sensitivity.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Organisms and appearance (Aspergillus niger, Candida).
- The essential role of **aural toilet**.
- Duration of treatment and what to do with a perforation.

## 5.2 Common pitfalls

COMMON PITFALLS

- Treating otomycosis with more antibiotic drops.
- Stopping clotrimazole as soon as symptoms settle.

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